

# Open-Ended Lab Task – Computer Networking

**Course:** Computer Networks

**Deadline:** 20 June 2026

## Scenario

A newly established educational institute is setting up a small computer network for its administration and academic departments. The institute requires a reliable and scalable network that can support communication between departments and provide future expansion capabilities.

## Task

1. Network Requirements Analysis: Identify networking needs and estimate the number of devices.
2. Topology Selection: Select a suitable network topology and justify your choice.
3. Network Cabling Plan: Recommend appropriate cable type(s) and explain your selection.
4. Switch Design: Determine the number of switches required, assign ports logically, and provide basic switch configuration commands where applicable.
5. Router Configuration: Design an IP addressing scheme, configure router interfaces, and demonstrate connectivity between network segments.
6. Network Diagram: Draw a complete logical network diagram showing PCs, switches, routers, connections, and IP addresses.
7. Future Improvements: Suggest at least three enhancements that can improve the network in the future.

## Deliverables

- Network requirements analysis
- Topology justification
- Cabling plan
- Switch configuration
- Router configuration
- IP addressing table
- Network diagram
- Future recommendations

## Evaluation Criteria

- Design correctness
- Creativity and justification
- Proper use of networking concepts
- Configuration accuracy
- Report presentation